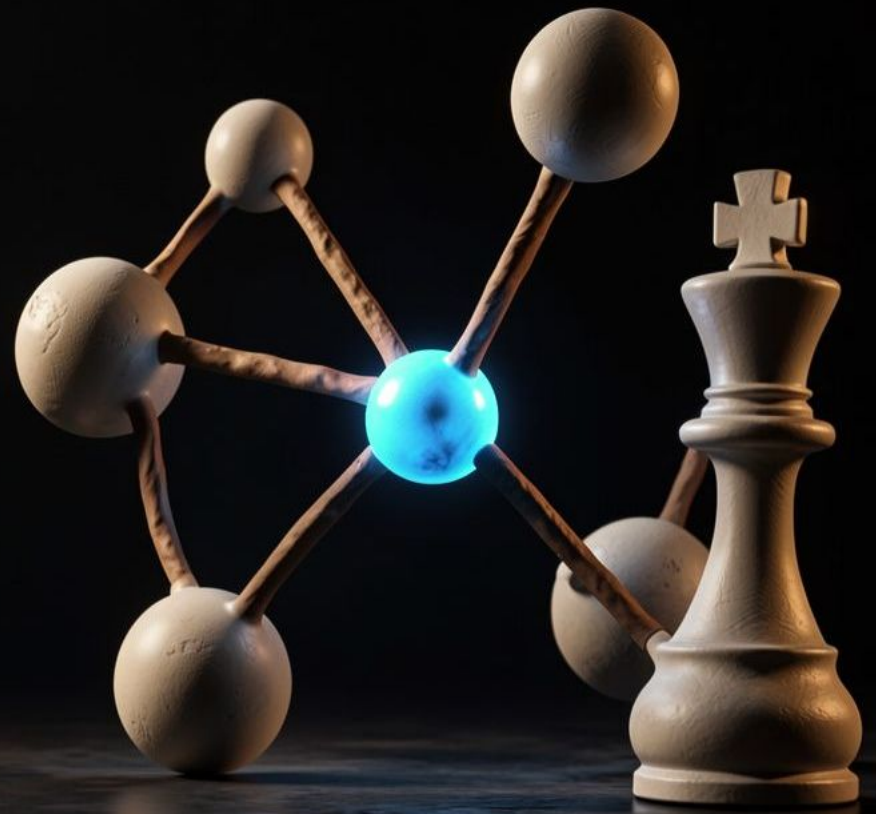
ALGORITHMS · GRAPH THEORY

Design and Analysis of Algorithms: Prim's and Kruskal's Algorithms

Minimum spanning trees through two greedy strategies

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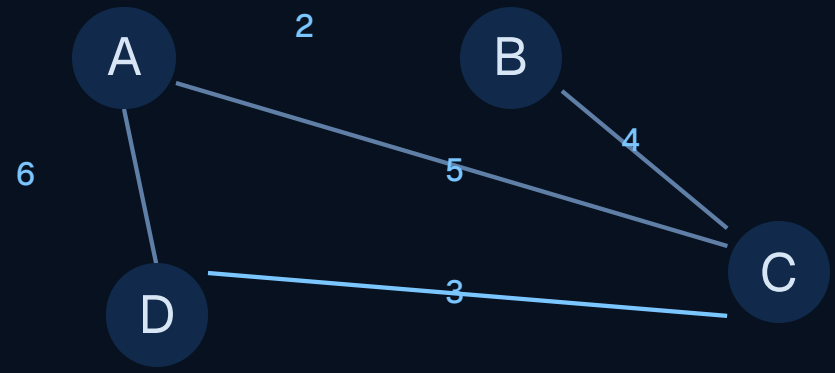
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01 · GRAPH VOCABULARY

Graphs turn connections into structure

- **Vertex:** a point such as A or B
- **Edge:** a connection between two vertices
- **Weighted graph:** each edge carries a cost or distance
- **Spanning tree:** connects every vertex with no cycle
- **Minimum spanning tree:** the lowest-cost spanning tree

WEIGHTED GRAPH



SPANNING TREE · 3 EDGES



Every vertex stays connected; no cycle.

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02 · THE TARGET

An MST connects everything at minimum total cost

V−1

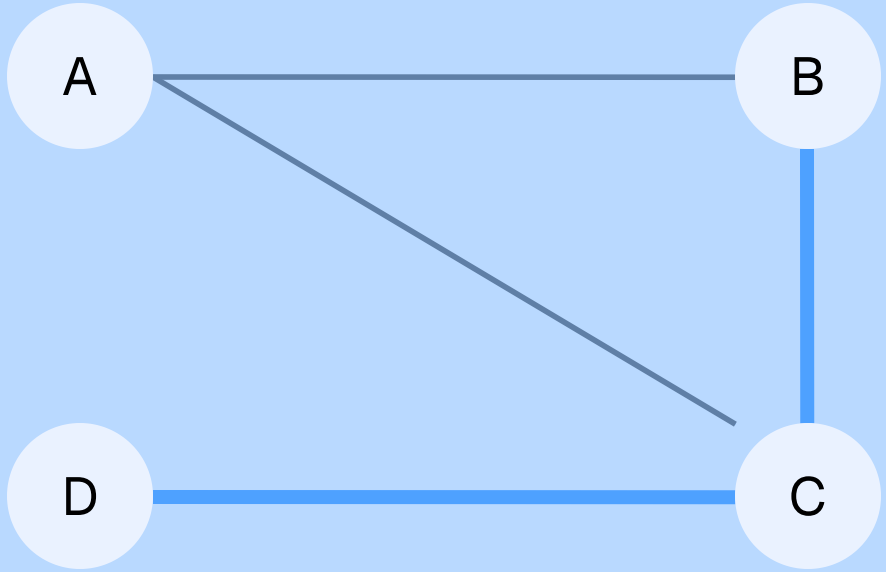
Edges for V vertices. Include every vertex.

Tree structure. Connected, with no cycles.

Minimum cost. Minimize the sum of selected edge weights.

Ties. More than one valid answer may exist.

WEIGHTED GRAPH · BLUE = TREE



Blue edges form the highlighted spanning tree: V vertices, V−1 edges.

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03 · PRIM

Prim's algorithm grows one tree from a starting vertex

The next edge is the cheapest safe connection leaving the current tree. The cut separates what is already connected from what is not.

CUT-BASED GREEDY CHOICE

01

Start

Start with any vertex.

02

Choose

Choose the lightest edge from the tree to an unvisited vertex.

03

Add

Add that edge and vertex.

04

Repeat

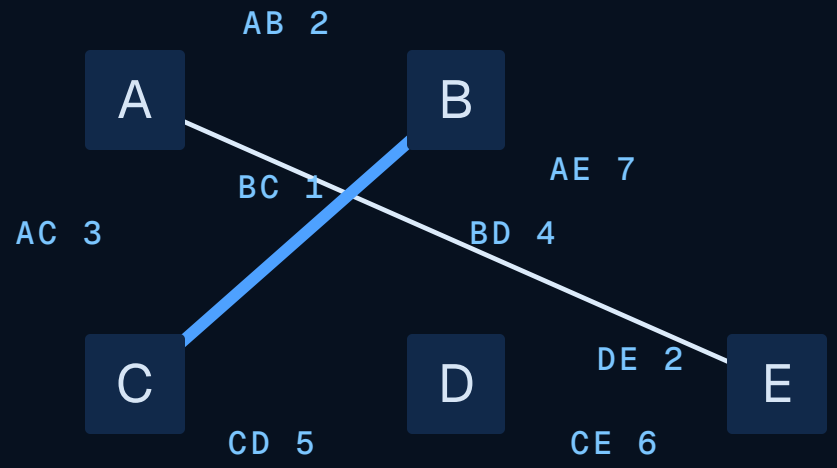
Repeat until every vertex is included.

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04 · WORKED EXAMPLE · PRIM FROM A

Prim's reaches weight 9 by expanding from A

Same weighted undirected graph throughout: AB=2, AC=3, BC=1, BD=4, CD=5, CE=6, DE=2, AE=7.



TREE GROWTH

1. Start {A}; select AB=2 → {A,B}
2. Select BC=1 → {A,B,C}
3. Select BD=4 → {A,B,C,D}
4. Select DE=2 → {A,B,C,D,E}

Final MST: BC, AB, BD, DE

9

Total weight = 1+2+4+2 = 9.

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05 · KRUSKAL

Kruskal's algorithm builds a forest, then joins it

Instead of growing from one start vertex, Kruskal considers every edge globally. It accepts an edge only when its endpoints are in different components.

1. Sort all edges by nondecreasing weight
2. Consider the next lightest edge
3. Accept it only if it does not create a cycle
4. Stop after V−1 accepted edges

SAFE-SET DECISION



Disjoint-set union tracks which vertices already belong together.

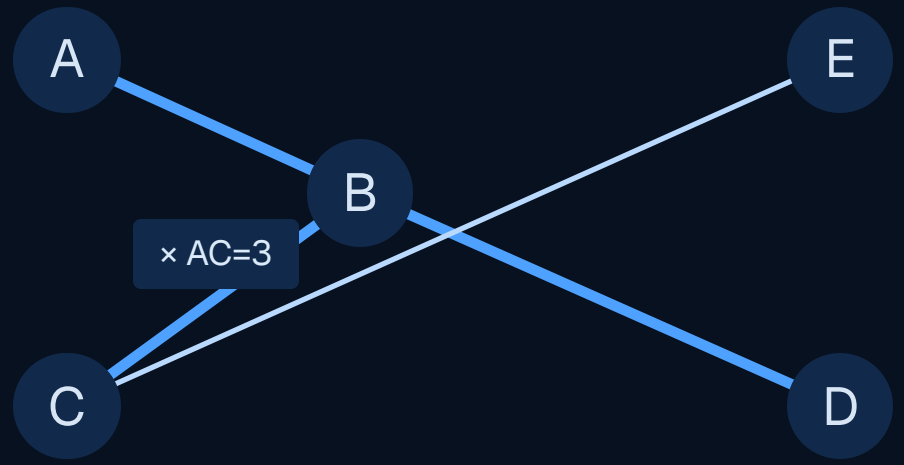
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06 · WORKED EXAMPLE · KRUSKAL

Kruskal accepts four edges and rejects one cycle

Sorted order: BC=1, AB=2, DE=2, AC=3, BD=4, CD=5, CE=6, AE=7.

SAME GRAPH · ACCEPTED EDGES IN BLUE



COMPONENTS

- **Accept:** BC=1, AB=2, DE=2
- **Reject:** AC=3; A, B, and C are connected, creating cycle A–B–C–A
- **Accept:** BD=4, joining the two components
- **Stop:** four edges connect five vertices

9

Total weight = 1 + 2 + 2 + 4

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07 · THE SHARED IDEA

Both algorithms rely on a greedy safe edge

Prim

```
Prim(G, start)
T ← ∅; S ← {start}
while |S| < |V|:
  choose minimum-weight edge (u,v)
  where u ∈ S and v ∉ S
  add (u,v) to T; add v to S
return T
```

Kruskal

```
Kruskal(G)
T ← ∅
sort E by nondecreasing weight
for each edge (u,v) in E:
  if Find(u) ≠ Find(v):
    add (u,v) to T
    Union(u,v)
  if |T| = |V|−1: break
return T
```

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Choose the algorithm that matches the graph representation

Prim

Approach: grows one connected tree
Edge selection: cheapest frontier edge
Cycle handling: frontier rule avoids cycles
Data structures: priority queue + adjacency list, or matrix
Time: $O(V^2)$ with matrix; $O(E \log V)$ with binary heap; $O(E + V \log V)$ with Fibonacci heap
Best fit: often suits dense graphs

Kruskal

Approach: merges a forest
Edge selection: globally lightest remaining edge
Cycle handling: DSU Find/Union detects cycles
Data structures: sorted edge list + disjoint set
Time: $O(E \log E)$ for sorting plus near-linear DSU work
Best fit: convenient for sparse graphs or edge lists

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09 · APPLICATIONS

MSTs connect real systems with less total cost

- **Network design:** connect sites with minimum cable or link cost
- **Road networks:** connect locations while minimizing construction length
- **Telecommunications:** design efficient backbone connections
- **Electrical distribution:** connect substations with minimum wiring cost



Prim grows outward from a seed. Kruskal selects globally and prevents cycles.

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Thank you

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